How-To Library

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Dynamic Data Masking Multi-Server Development and Deployment

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Abstract

This article describes an example architecture, development, and deployment of Dynamic Data Masking Multi-Server architecture.

Supported Versions

- Dynamic Data Masking 9.8.1 or later

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Overview

Dynamic Data Masking Multi-Server architecture refers to multiple Dynamic Data Masking Servers running in an enterprise environment. There are several ways to implement Dynamic Data Masking Multi-Server architecture. The architecture described in this article is based on an architecture implemented by a Dynamic Data Masking customer. You can use this example as a starting point to understand the possible variations of multi-server architecture. This architecture was designed based on the requirements of the customer and can be changed to fit the requirements of your organization.

This implementation of Dynamic Data Masking Multi-Server architecture consists of one Development Central DDM Server, one Production Central DDM Server, and multiple Development Deployed and Production Deployed DDM Servers. There are distinct development and production environments. Dynamic Data Masking configuration and testing takes place in the development environment. After testing is complete, you copy the configuration data into the production environment. To limit the number of people that have access to production data, the production environment is closed. The Dynamic Data Masking Management Console does not access the production environment and the database client does not access the Production Central DDM Server.

Implementation of the architecture includes the following steps:

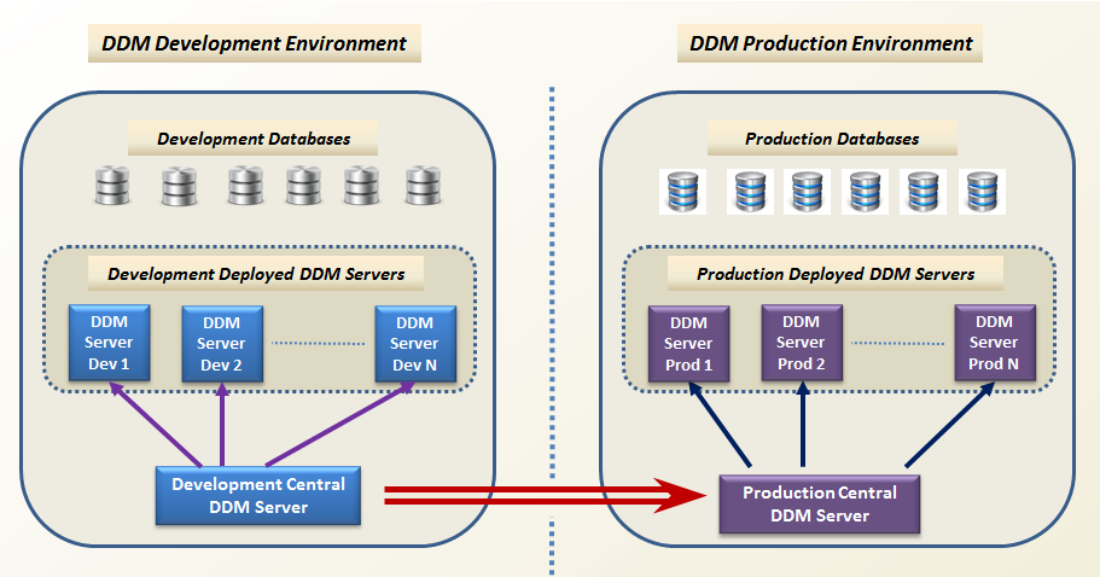
1. Create Dynamic Data Masking policies and rules on the Development Central DDM Server.
2. Import the Dynamic Data Masking configuration data from the Development Central DDM Server to the Development Deployed DDM Servers.
3. Perform testing on the Development Deployed DDM Servers.
4. Import the Dynamic Data Masking configuration data from the Development Central DDM Server to the Production Central DDM Server.
5. Import the Dynamic Data Masking configuration data from the Production Central DDM Server to the Production Deployed DDM Servers.

Architecture

This architecture consists of a development environment and a production environment with multiple deployed servers in each environment. Configuration and testing take place in the development environment. After testing, you copy the configuration data from the development environment to the production environment. The movement of configuration

data from the development environment to the production environment is indicated by the red arrow in the image below.

The following image shows the Dynamic Data Masking Multi-Server architecture:



The architecture includes the following components:

Development Central DDM Server

The Development Central DDM Server is the Dynamic Data Masking Server on which you create Dynamic Data Masking policies and rules. It contains definitions for the servers in the development and production environments.

The Development Central DDM Server is the only server that is accessible by the Management Console. It is not accessible by the database client.

You import the Dynamic Data Masking configuration data that you create on the Development Central DDM Server to the following locations:

- Development Deployed DDM Servers
- Production Central DDM Server

Development Deployed DDM Servers

The Development Deployed DDM Servers contain configuration data that you import from the Development Central DDM Server. The Development Deployed DDM Servers are accessible by the database client. You can use the servers for testing before you move the configuration data to the production environment.

Production Central DDM Server

The Production Central DDM Server contains Dynamic Data Masking configuration data that you import from the Development Central DDM Server. The Production Central DDM Server is not accessible by the Management Console or the database client.

Development Deployed DDM Servers

The Development Deployed DDM Servers contain configuration data that you import from the Production Central DDM Server. The servers are accessible by the database client.

Use the following guidelines when you implement the architecture:

- Because all development takes place on the Development Central DDM Server, the Development Central DDM Server contains definitions for development and production environments. You use the Check Server Name connection rule matcher to apply the connection rule action to the correct Dynamic Data Masking Server.
- The Dynamic Data Masking Server names on the development and production environments cannot be the same. Each server must have a unique name. You specify the server name in the Check Server Name connection rule matcher. You can use the server name to distinguish between the development and production environments.
- You must define database nodes in the Management Console for the development and production environments. Each database node must have a unique name.
- For each Dynamic Data Masking service, you must create connection rules that contain associations for development and production environments.
- You only need to define the security rule sets once. The security rule sets are identical in the development and production environments.

Deploying Multi-Server Architecture

To set up the multi-server architecture, you must complete the deployment on the development environment, the Production Central DDM Server, and the Production Deployed DDM Servers.

Deployment on the Development Environment

To complete the deployment of the development environment, perform the following steps for each Development Deployed DDM Server:

1. Perform the silent installation of the Dynamic Data Masking Server on the Development Deployed DDM Server.
2. Stop the Dynamic Data Masking Server.
3. Copy the config.properties file from the Development Central DDM Server to the Development Deployed DDM Server.
4. Copy the jar files that the security rule sets reference from the Development Central DDM Server to the Development Deployed DDM Server.
5. Rename the Dynamic Data Masking Server.
6. Start the Dynamic Data Masking Server.
7. Synchronize the Development Deployed DDM Server with the Development Central DDM Server. You can use the `config sync` Server Control command to sync the servers.

Deployment of the Production Central DDM Server

To complete the deployment of the Production Central DDM Server, perform the following steps:

1. Perform the silent installation of the Dynamic Data Masking Server on the Production Central DDM Server.
2. Stop the Dynamic Data Masking Server.
3. Copy the config.properties file from the Development Central DDM Server to the Production Central DDM Server.
4. Copy the jar files that the security rule sets reference from the Development Central DDM Server to the Production Central DDM Server.
5. Rename the Dynamic Data Masking Server.
6. Start the Dynamic Data Masking Server.

7. Synchronize the Production Central DDM Server with the Development Central DDM Server. You can use the `config sync` Server Control command to sync the servers.

Deployment of Production Deployed DDM Servers

To complete the deployment of the production environment, perform the following steps for each Production Deployed DDM Server:

1. Perform the silent installation of the Dynamic Data Masking Server on the Production Deployed DDM Server.
2. Stop the Dynamic Data Masking Server.
3. Copy the config.properties file from the Production Central DDM Server to the Production Deployed DDM Server.
4. Copy the jar files that the security rule sets reference from the Production Central DDM Server to the Production Deployed DDM Server.
5. Rename the Dynamic Data Masking Server.
6. Start the Dynamic Data Masking Server.
7. Synchronize the Production Deployed DDM Server with the Production Central DDM Server. You can use the `config sync` Server Control command to sync the servers.

Modifying the Dynamic Data Masking Configuration

You can modify the Dynamic Data Masking services, databases, and security rule sets. When you modify the Dynamic Data Masking configuration, you test the configuration on the Development Deployed DDM Servers before you import the configuration into the production environment.

Service Modifications

Complete the following steps when you modify the Dynamic Data Masking services:

1. Export the service from the Development Central DDM Server to a file. You can use the `service export` Server Control command to export the service. You can optionally save the file in a source control system for tracking purposes.
2. Import the service into each Development Deployed DDM Server. You an use the `service import` Server Control command with multiple targets to import the service.
3. Verify the service modifications on the Development Deployed DDM Servers.
4. Copy the service export file from the development to the production environment.
5. Import the service into the Production Central DDM Server. You can use the `service import` Server Control command to import the service.
6. Import the service from the Production Central DDM Server into each Production Deployed DDM Server. You can use the `service import` Server Control command with multiple targets to import the service.

Database and Security Rule Set Modifications

Complete the following steps when you modify database nodes and security rule sets:

1. Export the database or security rule set from the Development Central DDM Server to a file. You can use the `config export` Server Control command to export the object. You can optionally save the file in a source control system for tracking purposes.
2. Import the database or security rule set into each Development Deployed DDM Server. You can use the `config import` Server Control command with multiple targets to import the object.
3. Verify the database or security rule set modifications on the Development Deployed DDM Servers.
4. Copy the database or security rule set export file from the development to the production environment.

5. Import the database or security rule set to the Production Central DDM Server. You can use the `config import` Server Control command to import the object.
6. Import the database or security rule set to each Production Deployed DDM Server. You can use the `config import` Server Control Command with multiple targets to import the object.

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